

Kubernetes-IaC-Deployment — Cheat Sheet

Chain C · Project C-4 · dense one-page recap — pairs with docs/interactive/index.html

1 · Terraform

Command	Needs real AWS creds?
<code>terraform fmt -check</code>	No
<code>terraform init -backend=false</code>	No
<code>terraform validate</code>	No
<code>terraform plan / apply</code>	Yes — not run in this project

- Terraform provisions the CLUSTER only — workloads are a separate kubectl/helm layer

2 · Probes + Rollout

Probe fails	Kubernetes does
readinessProbe	Removes from Service endpoints (no restart)
livenessProbe	Kills + restarts the container

- `maxSurge: 1`, `maxUnavailable: 0` = zero-downtime rolling update

3 · Helm

- `helm lint` — checks chart STRUCTURE only
- `helm template | kubeconform` — validates RENDERED output (the one that actually matters)
- `helm upgrade --set image.tag=<sha>` — override without editing the chart

4 · GitOps (Flux)

Resource	Role
GitRepository	Polls a git url/ref, produces an artifact
Kustomization	Applies raw manifests from a path, with pruning
HelmRelease	Reconciles a Helm chart, like <code>helm upgrade --install</code> on a timer
ImageRepository / ImagePolicy	Image automation — writes new tags back to git

- CI = push model. Flux = pull model. Same end state, opposite initiator.

This project's actual (static) verification

Tool	What it checked
<code>terraform validate</code>	HCL internal consistency
<code>kubeconform -strict</code>	Real K8s manifests against official OpenAPI schema, offline
<code>kubeconform</code> + datreeio CRD catalog	Flux custom resources (GitRepository/Kustomization/HelmRelease)
<code>helm lint</code> + <code>helm template</code>	Chart structure + rendered output
<code>actionlint</code>	GitHub Actions workflow syntax